APPLIED OPTICS & PHOTONICS CONCENTRATION

(4) Applied Physics Core Courses + (1) ECE Core Course + (3) 700-level Electives + (2) Additional Electives

CORE COURSES
Applied Physics

Select (4) of the following

615.641 * Mathematical Methods for Physics & Engineering
Every Term • Asynchronous

615. 642 Electromagnetics
FA / SP • Asynchronous

615.653 Classical Mechanics
Every Term • Asynchronous

615.654 Quantum Mechanics
FA / SP • Asynchronous

615.665 Modern Physics
SP / SU • Asynchronous

615.671 Principles of Optics
Every Term • Asynchronous

Unshaded boxes indicate elective courses not offered by the Applied Physics program. Check the EP website / SIS for course offering information.

CORE COURSES
Electrical & Computer Engineering

Select (1) of the following

525.613 Fourier Techniques in Optics

525.625 Laser Fundamentals

525.691 Fundamentals of Photonics

* Students interested in taking 615.653/615.654 must complete 615.641 as a prerequisite

ELECTIVES

525.613 Fourier Techniques in Optics

525.625 Laser Fundamentals

525.636 Optics & Photonics Lab

525.674 Image Processing and Analysis

525.691 Fundamentals of Photonics

Select at least (3) of the following 700-level courses

615.751 Modern Optics
Fall • Asynchronous

615.778 Optical System Design and Modelling
Every Term • Asynchronous

615.780 Optical Detectors and Applications
FA • Asynchronous

615.781 Quantum Information Processing
SP • Asynchronous

716.782 Optics and MATLAB
SP • Asynchronous

525.753 Laser Systems and Applications

525.756 Optical Propagation, Sensing, and Backgrounds

525.772 Fiber Optic Communication Systems

525.797 Advanced Fiber Optic Laboratory

525.796 Introduction to High-Speed Optoelectrics

585.734 Biophotonics

Test your degree path with the Applied Physics Course Mapping Tool